Architecture, Traceability, and Technical Verification of a Web and Mobile Platform for Restaurant Operations: The SaborExpress Case Study

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Abstract—This paper examines which claims about SaborExpress integration can be supported by versioned repository evidence and reported test executions. A descriptive documentary case study combines targeted literature surveillance, contract inspection, and a review of published validation records. SaborExpress integrates an ASP.NET Core 8 API, a Next.js web client and an Expo/React Native mobile client supporting restaurant ordering, kitchen, cashier, payment and delivery workflows. A November 3, 2026 documentation report records 90 security, 62 QA and 11 PDF-invoice local backend tests passed on a candidate SHA, along with 7 web and 8 mobile tests on distinct versions. These outcomes concern limited test suites and do not establish overall acceptance. Candidate backend readiness returned HTTP 503 locally because migrations were pending, whereas an earlier staging SHA had passed its smoke checks. The case supports architectural characterization and partial validation but not causal improvements, full production readiness or user acceptance. Reproducible evaluation instruments and remaining gaps are presented.

Index Terms—restaurant information systems; software engineering; case study; traceability; validation; web and mobile applications

I. INTRODUCTION

Restaurant operations integrate catalogues, branches, employees, orders, kitchens, payments and deliveries. Changes in backend permissions or state contracts may affect independent web and mobile clients. Food-delivery literature investigates trust, perceived quality and user experience [1]–[5], and technology adoption models offer variables for future field work [6], [7]. These publications do not constitute evidence of SaborExpress outcomes. This study explicitly distinguishes declared architecture, observed testing, outstanding validation and evidence limitations.

II. RESEARCH PROBLEM AND OBJECTIVES

Earlier documentation identified plausible risks involving order consistency, payments, branch isolation and delivery coordination, but did not establish a baseline from an actual restaurant. The analytical problem tree treats insufficiently demonstrated operational traceability as its central problem; hypothetical causes include inter-client contract drift, incomplete cross-branch authorization coverage and dependencies on third-party payment services. Effects such as inconsistent states remain hypotheses, not measured incidents. Research question: Which integration capabilities and technical verification results can be reproducibly identified in SaborExpress, and what evidence is missing before operational readiness can be claimed? The objectives are to identify components, relate risks to literature and standards, report results by SHA and environment, and design reproducible functional and user-study instruments.

III. RELATED WORK AND TECHNOLOGY SURVEILLANCE

Shroff et al. [1] and Shankar et al. [2] survey the online food-delivery domain. Prasetyo et al. [3], Fakfare [4] and Koay et al. [5] examine satisfaction and service attributes in settings distinct from this project. Davis [6] and Venkatesh et al. [7] provide constructs for a future acceptance study rather than measured results here. Kruchten [8] supports architectural views and Taibi et al. [9] provides comparative patterns without suggesting that a modular backend must be a microservice system. Yin [10] informs case-study evidence; Kitchenham and Charters [11] inform transparent source selection without making this a registered systematic review. ISO/IEC 25010 [12], OWASP [13], and HTTP/JWT specifications [14], [15] inform quality and security criteria. The companion surveillance matrix contains 15 sources, identifiers, summaries and limitations.

IV. METHODS AND INSTRUMENTS

This is a descriptive, documentary, non-experimental case study updated to November 3, 2026. The bounded system includes SaborExpressBack, SaborExpressWeb, SaborExpressMovil and SaborExpressDocs. Documentary instruments are the technology extraction record, static API-client contract table, and assessment of the 9-traceability/CIERRE-HU-2026-11-03.md report. Evidence is classified into executed checks linked to a version, static implementation findings, and unexecuted proposals. No participants were recruited or interviewed in preparing this revision. Reproduction requires repository SHA, runtime environment, commands and CI artefacts. Personal data, secrets and actual payment transactions are excluded.

V. OBSERVED RESULTS

SaborExpress documents a central ASP.NET Core 8/EF Core backend with JWT authentication and SignalR, a Next.js administrative interface and an Expo/React Native mobile app. It covers branches, catalogue, employees, orders, kitchen operations, payments and deliveries. A previously generated OpenAPI snapshot reports 269 routes, 346 operations and 84 schemas; these counts are not runtime validation. Static route-and-method comparison found 92/92 matching Web signatures and 47/47 Mobile signatures against the earlier backend OpenAPI; payload and behavior contracts were not tested. The backend candidate SHA 17a8ce4 reports 90/90 Security, 62/62 QA and 11/11 Invoice PDF local tests passing. Web SHA aa5cb8d reports 7/7 tests and Mobile SHA af52bba reports 8/8 tests. A local web Playwright check used synthetic API data and passed a single test spanning 11 screens; it is not a user study. Candidate backend local /health returned 200 and /health/ready returned 503 owing to pending migrations. An earlier staging SHA ac2b39e passed smoke verification on October 29, 2026, but did not validate the candidate. Historical k6 evidence included one run satisfying thresholds and two failing the Server-Timing header requirement. These are extracted documentary results, not tests re-executed to create this paper.

VI. DISCUSSION AND VALIDITY

Unit and CI tests establish bounded behavior, not complete production workflows. The latest backend candidate remained outside main, while web and mobile records represent other versions. Static route matches do not establish authorization, idempotent processing or concurrent consistency. OWASP cross-branch checks, Wompi event signature and replay protection,

reconnection, secret rotation and full seven-role journeys require further execution. Construct validity is limited because route declarations differ from observed behavior. Internal validity is limited by the lack of a before-and-after experiment. External validity is restricted to this case. Reproducibility depends on immutable SHAs, execution records and controlled environments. No actual restaurant interview or user acceptance survey is reported.

VII. FUTURE VALIDATION PROTOCOL

A field pilot should recruit an authorized operational representative and users with informed consent. The researcher should record branch, role, task, version, observation date and anticipated result; apply a semi-structured interview and a task observation sheet; anonymize responses and preserve failures. A separate isolated test environment should cover seven roles across two branches, forbidden object access, duplicate or modified payment webhook events in sandbox, network reconnection, declared load-test thresholds, migration rollback, recovery and physical-device/printing validation. Companion templates are ready for use but have not been applied.

VIII. CONCLUSIONS

The study documents a multi-client restaurant platform and a traceability chain linking source code, standards and versioned validation outcomes. Local tests and CI artefacts offer real evidence beyond static inspection, but are insufficient for claims of user acceptance, complete production readiness, or closure of every user story. The research contribution is a reproducible distinction between observed implementation, bounded verification and the field and technical evaluation needed to test the remaining hypotheses.

IX. DECLARATIONS

Manuscript author named: Yerson Stiven Cuéllar Rubiano, subject to personal confirmation of authorship and contribution. Affiliation: SENA, Centro de la Industria, la Empresa y los Servicios, ADSO cohort 3145555. Funding and conflicts: not established in the supplied record; the author must confirm them before submission. Ethics: no participants were enrolled in preparing this revision; future participant studies require consent and institutional authorization. Evidence: cited public repositories and their versioned reports. Artificial intelligence: ChatGPT (OpenAI, GPT-6 family) assisted on November 3, 2026 with restructuring, drafting, translation and document generation. Reported test outcomes were extracted from existing project evidence, not experimentally generated by the assistant. The human author must verify references, review claims and approve the declaration; any additional tools used must also be disclosed.

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